```
r1.low, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood
     1 20263016             5000           0.277             -37300.0
     2 20263017             5000           0.276             -37287.8
     3 20263018             5000           0.277             -37292.3
     4 20263019             5000           0.275             -37291.7
 best.log.likelihood errored.proposals
            -37275.2                 0
            -37278.3                 0
            -37278.7                 0
            -37277.0                 0
                                   file
 mcmc_r1.low_step2_chain1_2026-10-19.Rdata
 mcmc_r1.low_step2_chain2_2026-10-19.Rdata
 mcmc_r1.low_step2_chain3_2026-10-19.Rdata
 mcmc_r1.low_step2_chain4_2026-10-19.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
                  hiv.specific.mortality.rates.0
                                           1.971
                       female.to.male.multiplier
                                           1.936
                 male.hiv.mortality.multiplier.0
                                           1.830
                                         trate.0
                                           1.565
          age.25.to.49.hiv.mortality.multiplier.0
                                           1.367
           age.20.to.29.transmission.multiplier.0
                                           1.334
      age.65.to.79.mortality.slope.multiplier.male
                                           1.317
  age.45.to.64.mortality.intercept.multiplier.male
                                           1.290
           age.0.to.14.hiv.mortality.multiplier.1
                                           1.287
                                age.assortativity
                                           1.263
           age.15.to.19.transmission.multiplier.3
                                           1.246
      age.45.to.64.mortality.slope.multiplier.male
                                           1.233
                                         trate.3
                                           1.230
           age.30.to.39.transmission.multiplier.3
                                           1.229
                         log.OR.suppression.slope
                                           1.201
           age.20.to.29.transmission.multiplier.1
                                           1.199
           age.30.to.39.transmission.multiplier.0
                                           1.186
         age.50.and.over.transmission.multiplier.3
                                           1.179
  age.65.to.79.mortality.intercept.multiplier.male
```

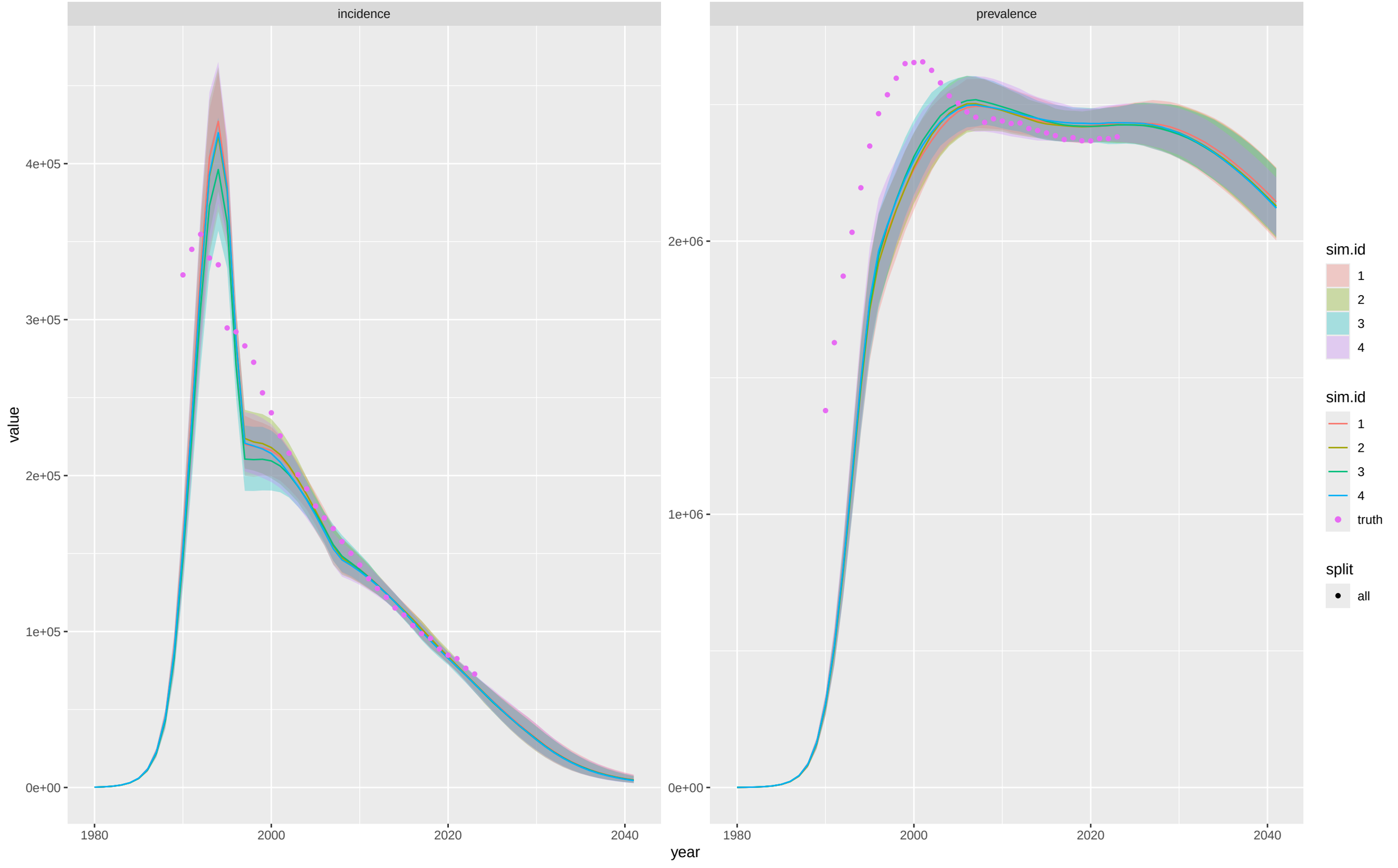
r1.low step 2: likelihood

0
−10000
−20000
−30000
value
0
25000
50000
75000
100000
iteration
chain
1
2
3
4
type
Log Likelihood
Log Prior
Log Prior x Likelihood

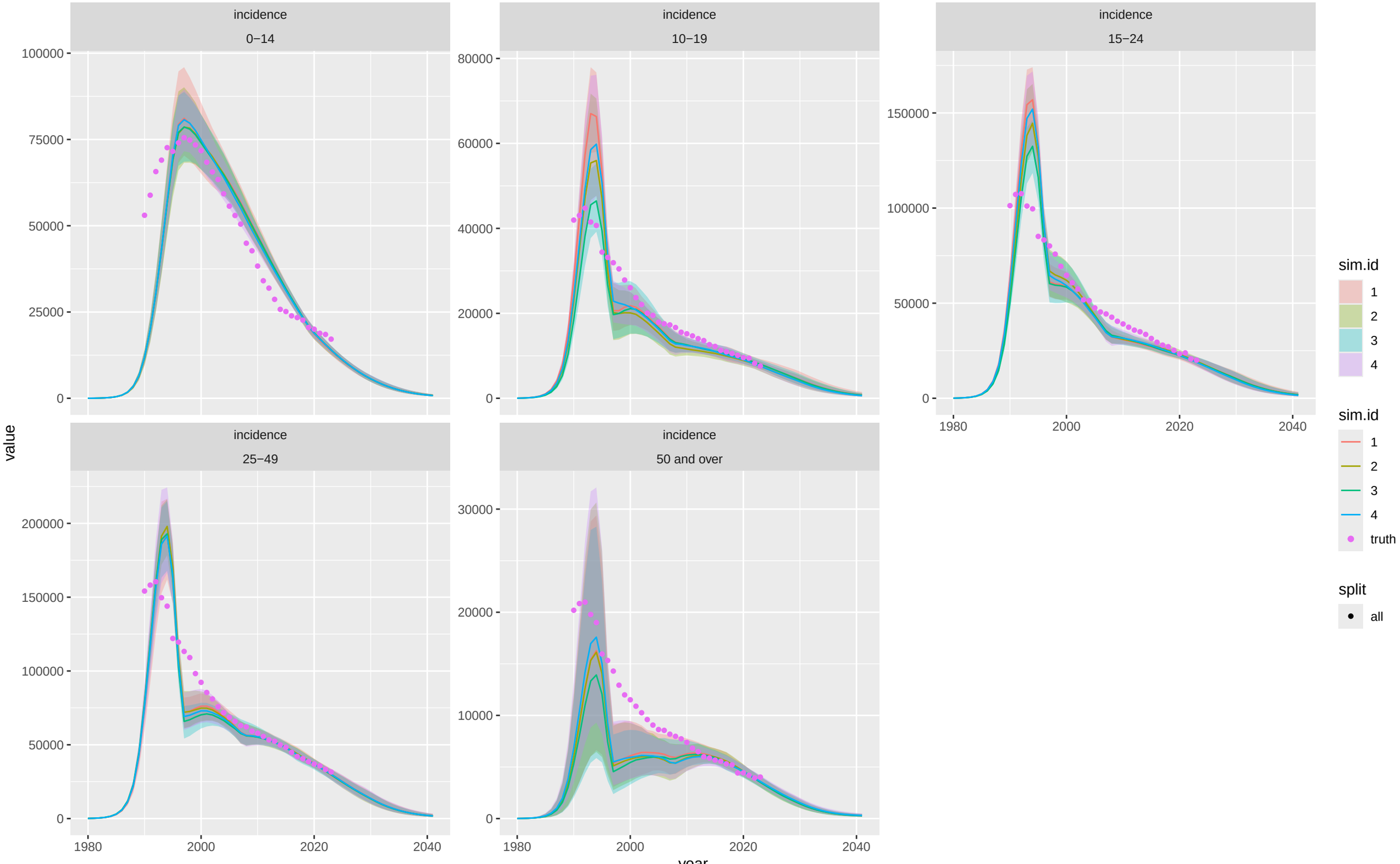
r1.low step 2: trace



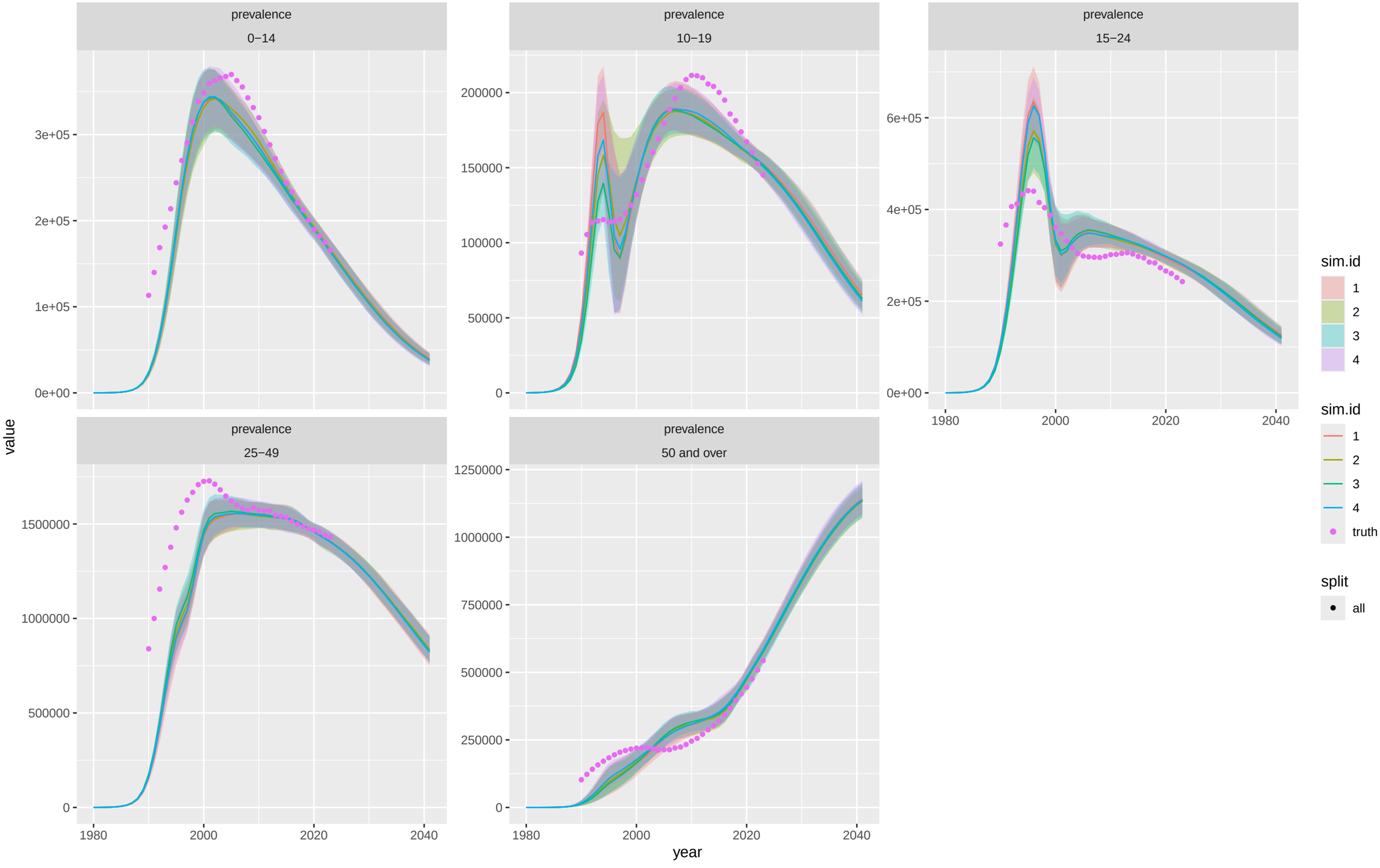
r1.low step 2: fit (one color per chain)



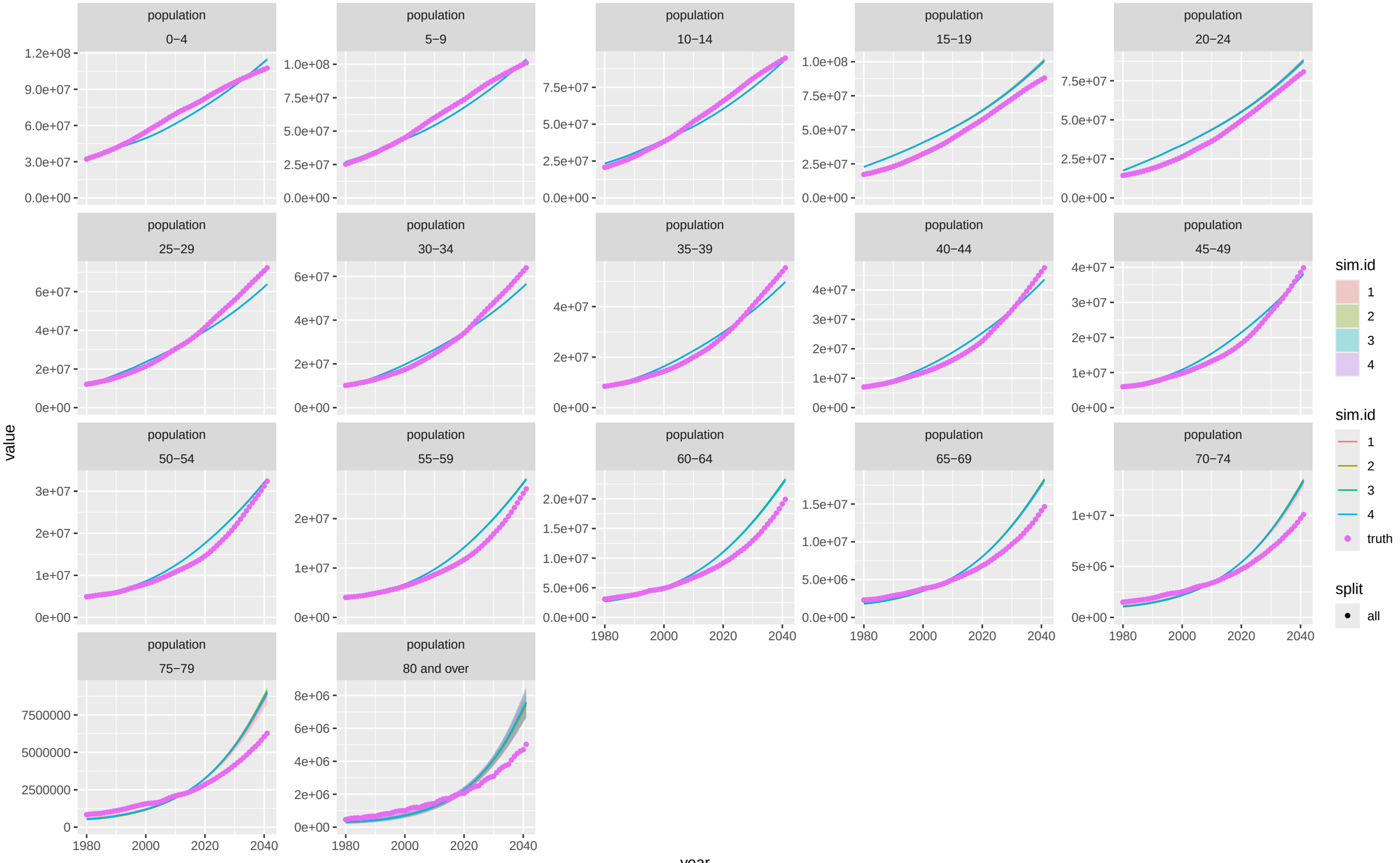
r1.low step 2: incidence.by.age (one color per chain)



r1.low step 2: prevalence.by.age (one color per chain)



r1.low step 2: population.by.age (one color per chain)



r1.low step 2: cascade (one color per chain)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

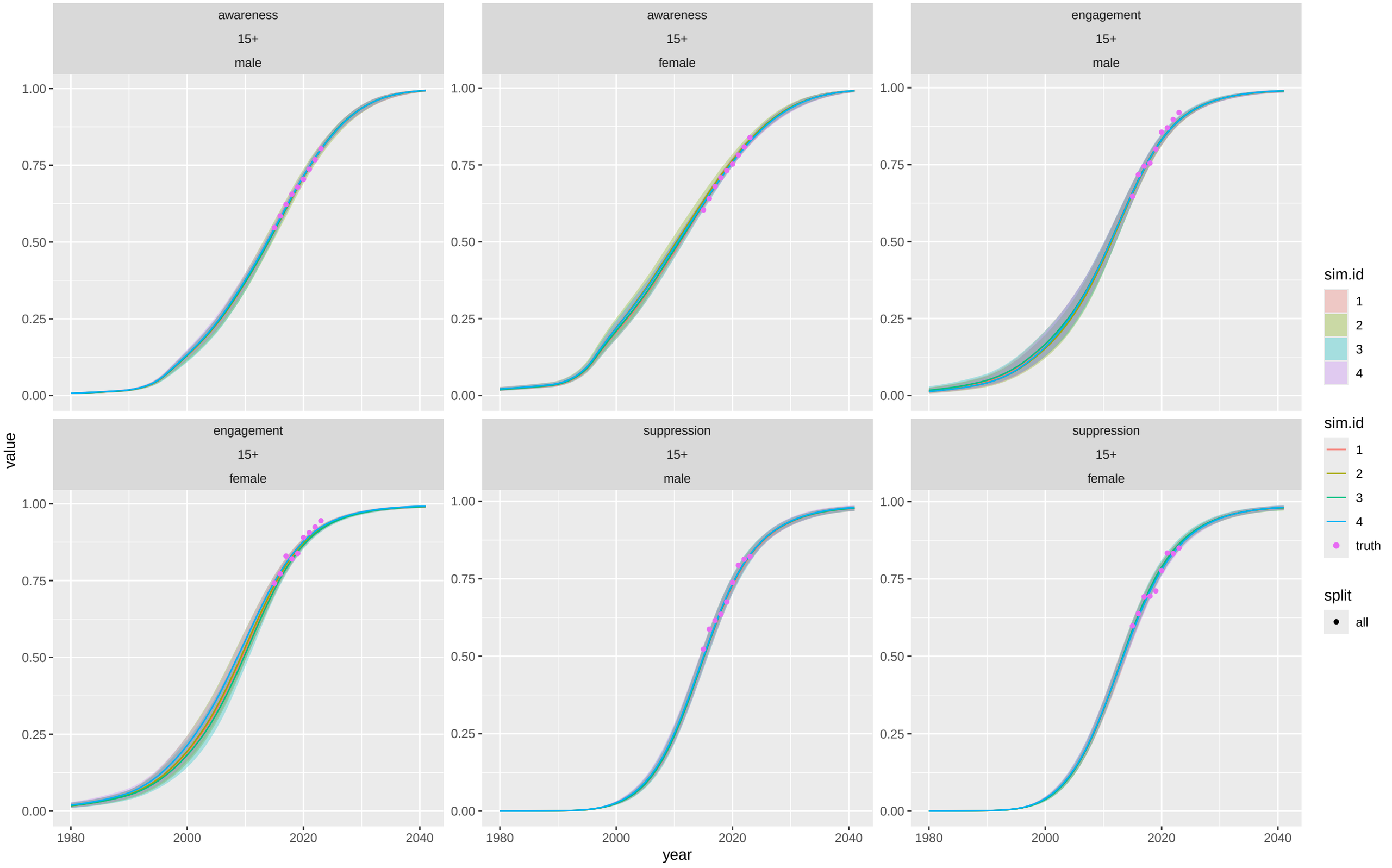
1980
2000
2020
2040

sim.id
1
2
3
4

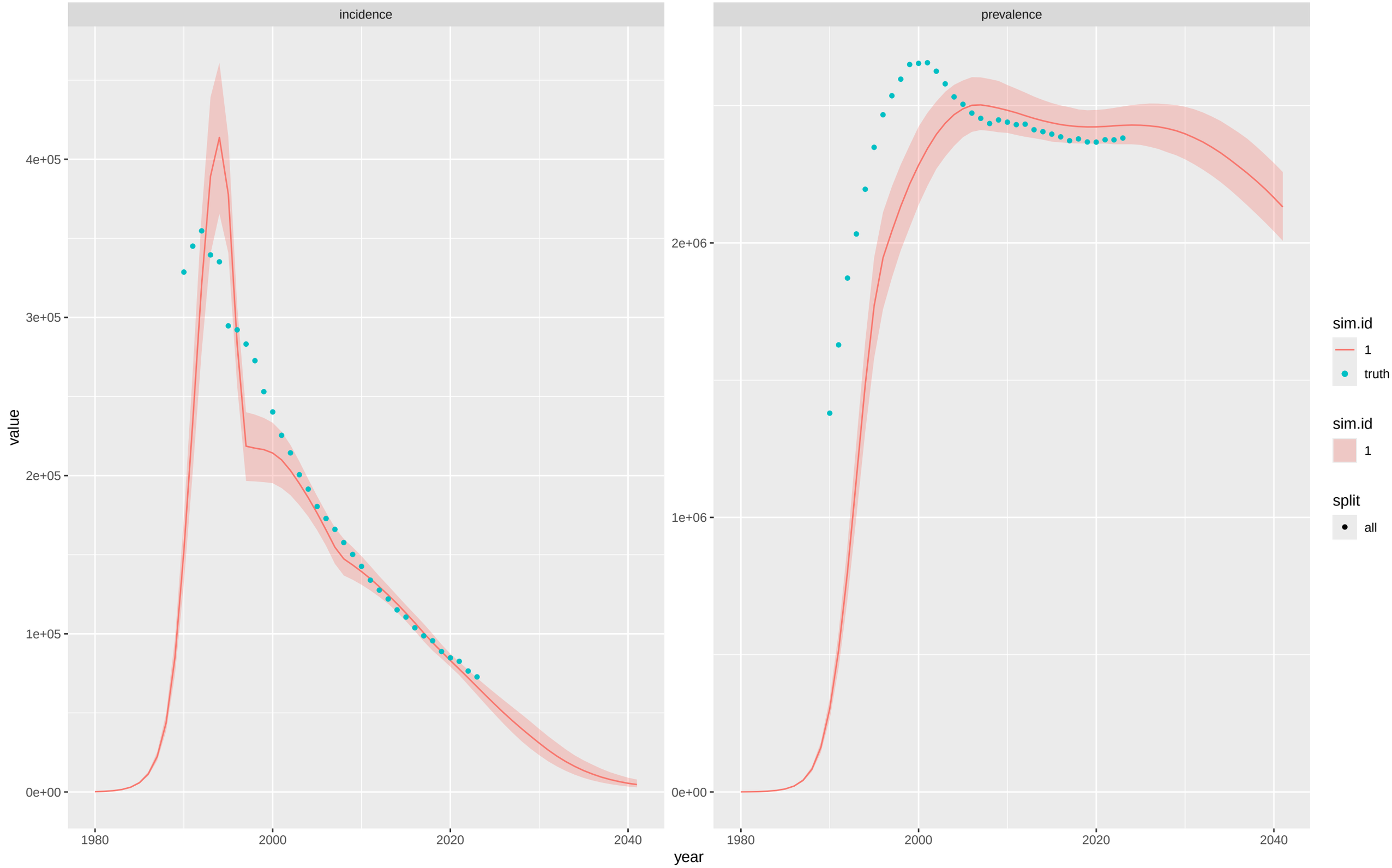
sim.id
1
2
3
4
truth

split
all

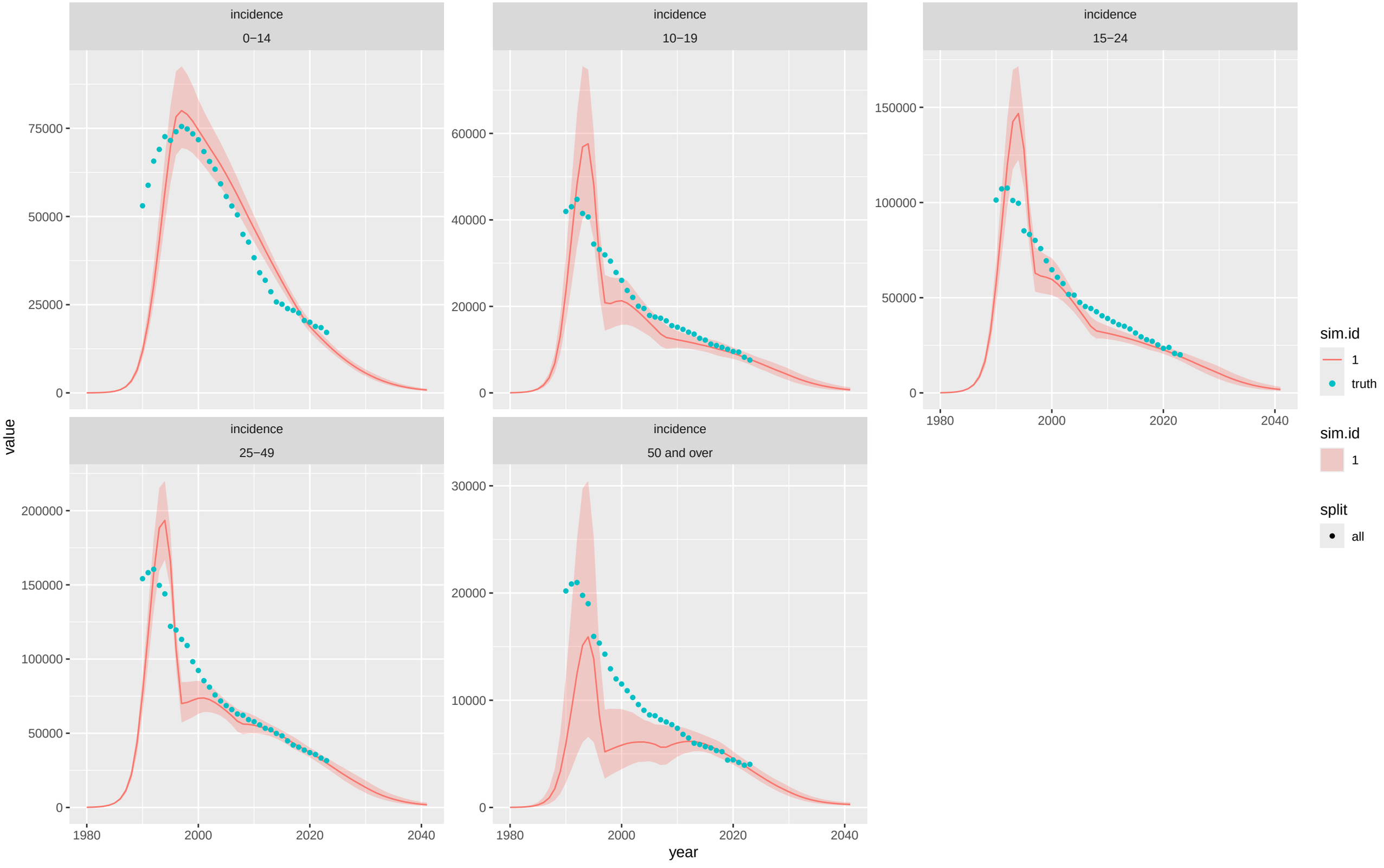
r1.low step 2: cascade.by.age.sex (one color per chain)



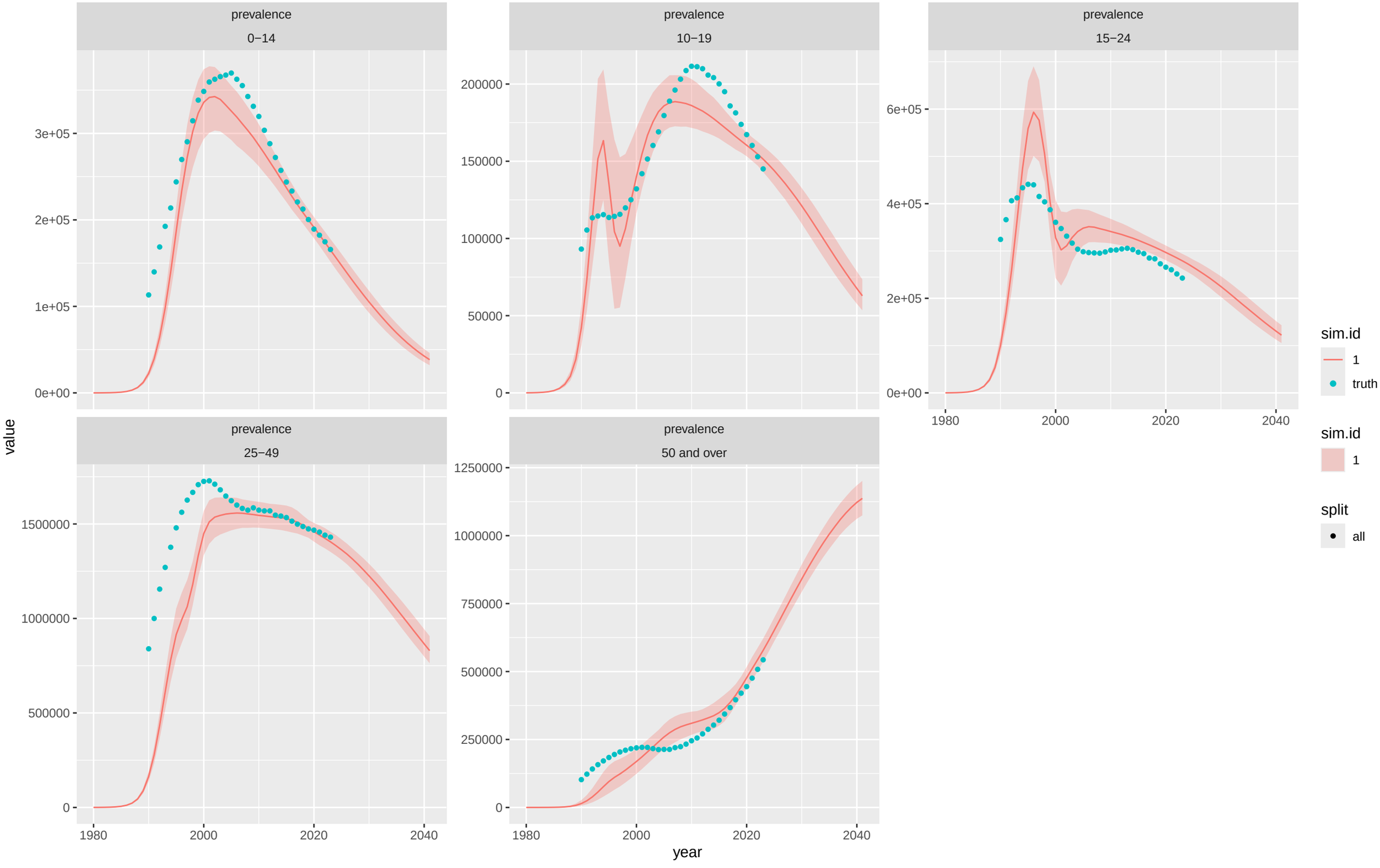
r1.low: fit (merged Step 2 simset)



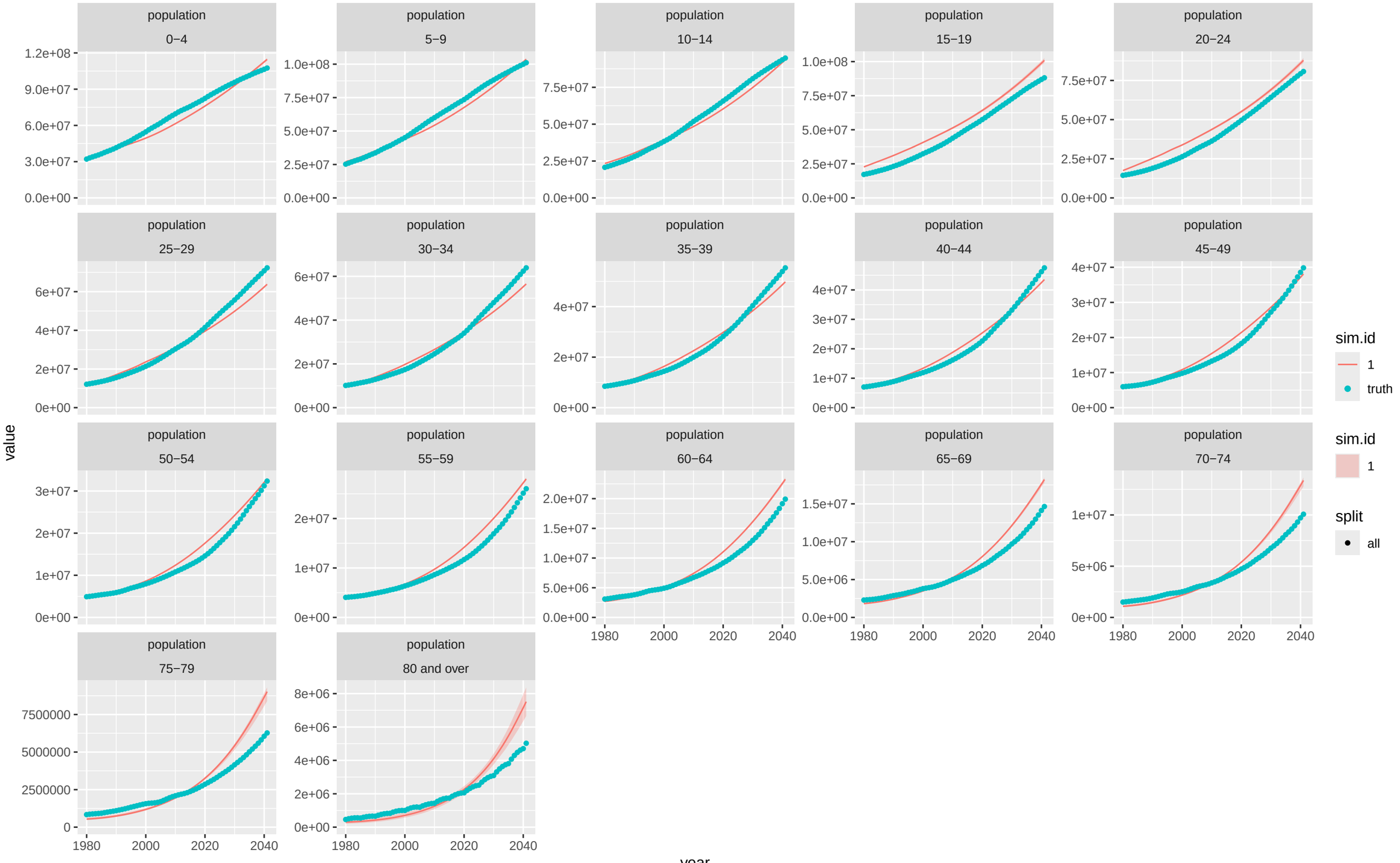
r1.low: incidence.by.age (merged Step 2 simset)



r1.low: prevalence.by.age (merged Step 2 simset)



r1.low: population.by.age (merged Step 2 simset)



r1.low: cascade (merged Step 2 simset)

awareness

engagement

suppression

1.00

0.75

0.50

0.25

0.00

value

1980

2000

2020

2040

year

sim.id

1

truth

sim.id

1

split

all

r1.low: cascade.by.age.sex (merged Step 2 simset)

awareness
15+
male

awareness
15+
female

engagement
15+
male

engagement
15+
female

suppression
15+
male

suppression
15+
female

value

1.00
0.75
0.50
0.25
0.00

1980
2000
2020
2040

year

sim.id
1
truth

sim.id
1

split
all